

MIDDLETON ISLAND, AK, USA

WMO: 703430

Lat: 59.433N

Lon: 146.333W

Elev: 31

StdP: 100.96

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25402

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-6.2	-4.1	-13.8	1.1	-4.5	-12.3	1.3	-3.2	21.7	2.5	19.7	3.0	6.8	20	0.798

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.1	17.4	14.1	16.2	13.5	15.0	12.9	14.9	16.9	14.1	16.1	13.4	15.0	3.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	9.9	15.6	13.0	9.3	14.7	12.5	9.1	14.4	41.4	16.8	39.5	16.0	37.7	14.9	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.1	14.1	DB	-7.1	18.8	2.4	1.9	-8.8	20.2	-10.2	21.3	-11.5	22.4	-13.3	23.8	(4)
(5)			WB	-8.8	15.4	2.2	1.6	-10.4	16.6	-11.7	17.5	-13.0	18.5	-14.6	19.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	2.3	2.2	2.2	4.7	7.1	10.3	12.6	13.1	11.1	7.3	4.0	2.8
	DBStd	4.62	2.91	2.67	2.59	1.85	2.49	1.65	1.55	1.44	1.71	2.12	2.76	2.45
	HDD10.0	1457	239	219	242	158	92	14	0	0	7	84	178	223
	HDD18.3	4251	497	452	501	408	347	241	177	162	216	341	428	482
	CDD10.0	248	0	0	0	1	3	23	81	96	42	2	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.1	7.5	7.2	6.7	6.3	5.5	4.5	4.1	4.3	5.5	6.6	7.3	7.8
Precipitation	PrecAvg	1950	188	157	165	140	120	85	111	157	215	237	190	185
	PrecMax	2388	383	268	567	233	247	127	166	319	351	417	439	383
	PrecMin	1202	31	62	90	70	40	35	44	39	74	100	62	75
	PrecStd	233	73	56	85	49	51	23	25	59	65	85	82	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	7.4	7.7	12.1	14.1	17.4	18.8	20.0	17.1	12.4	9.2	7.9
		MCWB	6.0	6.0	5.7	8.4	10.1	13.6	15.5	16.2	12.9	10.6	8.0	6.4
	2%	DB	7.1	6.4	7.0	8.9	12.6	15.2	17.3	17.6	15.2	11.3	8.7	7.4
		MCWB	5.6	4.8	5.2	6.3	9.6	12.3	14.3	14.5	12.6	9.7	7.8	6.1
	5%	DB	6.3	6.1	6.2	7.8	11.3	13.8	16.1	16.2	14.0	10.9	7.9	6.9
		MCWB	4.8	4.6	4.5	5.5	8.8	11.3	13.8	13.8	11.9	9.5	6.6	5.5
	10%	DB	6.0	5.2	5.6	7.3	10.0	12.7	14.8	15.0	13.0	9.9	7.5	6.2
		MCWB	4.6	3.9	4.0	5.1	8.1	10.8	12.9	13.2	11.3	8.7	6.0	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	6.2	6.7	8.8	11.0	14.3	15.8	16.8	14.0	11.3	8.9	7.8
		MCDB	7.3	7.0	7.3	11.9	13.8	16.9	18.1	19.4	15.6	11.7	9.3	8.5
	2%	WB	6.0	5.2	5.7	6.7	10.1	12.8	14.7	15.3	13.2	10.2	8.0	6.8
		MCDB	6.9	6.3	6.8	8.2	12.4	15.1	16.8	17.3	14.7	10.9	8.2	7.3
	5%	WB	5.3	4.6	4.8	5.8	9.1	11.8	14.1	14.4	12.6	9.5	7.5	6.0
		MCDB	6.4	5.9	6.2	7.4	11.0	13.2	16.1	16.2	13.9	10.3	7.9	6.9
	10%	WB	4.7	4.0	4.0	5.2	8.1	11.0	13.3	13.6	12.0	8.7	6.7	5.1
		MCDB	5.9	5.3	5.4	7.0	9.5	12.6	14.9	15.1	13.1	9.5	7.6	6.3
Mean Daily Temperature Range	5% DB	MDBR	2.6	2.6	3.2	3.6	3.6	3.5	3.2	3.1	3.2	2.7	2.7	2.7
		MCDBR	2.7	2.8	3.6	4.8	5.6	5.0	5.0	4.7	4.2	3.2	2.6	3.0
		MCWBR	2.9	2.8	3.1	3.5	3.6	3.4	3.4	3.3	3.2	2.7	2.8	3.0
	5% WB	MCDBR	2.8	2.7	3.3	4.7	5.3	4.5	4.9	4.7	3.7	2.8	2.4	2.9
		MCWBR	3.0	2.8	3.1	3.4	3.5	3.3	3.4	3.3	3.1	2.8	2.8	3.1
Clear-Sky Solar Irradiance	taub	0.280	0.313	0.327	0.367	0.373	0.368	0.371	0.368	0.352	0.329	0.287	N/A	
	taud	2.257	2.390	2.429	2.321	2.372	2.451	2.481	2.489	2.485	2.383	2.223	N/A	
	Ebn at Noon	526	710	817	833	855	866	854	830	781	671	494	N/A	
	Edh at Noon	42	63	80	106	108	102	97	90	77	62	42	N/A	
All-Sky Solar Radiation	RadAvg	0.28	0.86	2.17	3.76	4.80	5.11	4.44	3.43	2.46	1.16	0.40	0.18	
	RadStd	0.04	0.16	0.26	0.37	0.45	0.41	0.59	0.48	0.35	0.11	0.06	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page